PROJECT TITLE: SWIFT SHIP TRACKER

Team lead: Vishnu prasath KM

Team member 1: Vishnu R

Team member 2: Marimuthu E

Team member 3: Mohammed Haarish M

Team member 4: Maga yoges

Phase 1: Requirement Analysis & Planning

1.1 Defining Project Scope & Core Objectives

- **Scalable Parcel Booking:** Support end-to-end parcel lifecycle management: `Booked` $\rightarrow$ `Dispatched` $\rightarrow$ `In Transit` $\rightarrow$ `Out for Delivery` $\rightarrow$ `Delivered` / `Delayed`.
- **Automated Customer Visibility:** Eliminate manual status updates by dispatching automated Email Alerts and in-app updates upon every status transition.
- **Agent Operational Workspace:** Provide dispatchers and delivery agents with centralized Lightning console pages and guided Flows for instant updates.
- **Enterprise Security Architecture:** Enforce strict data privacy using Salesforce Profiles, Permission Sets, and Role Hierarchies to safeguard customer addresses and courier routes.
- **Conversational AI Tracking:** Embed an autonomous AI agent capable of parsing natural language tracking requests and retrieving real-time status data without human intervention.

1.2 Stakeholder & User Persona Matrix

- **Customer / Sender:**
  - *Responsibilities:* Books parcel shipments, inputs destination addresses, monitors delivery progress, and receives real-time delivery notifications.
  - *Touchpoints:* Self-service portal (Experience Cloud), transactional emails, and the conversational AI chat interface.
- **Delivery Agent:**
  - *Responsibilities:* Reviews daily assigned parcel runs, updates checkpoint locations, and confirms physical customer handover.
  - *Touchpoints:* SwiftShip Tracker Lightning Console, Quick Actions, and mobile-responsive status buttons.
- **Customer Support Team:**
  - *Responsibilities:* Handles delivery exception escalations, re-routes delayed packages, and resolves customer support cases.
  - *Touchpoints:* Service Cloud Case feed, Omni-Channel routing, and parcel relationship records.
- **System Administrator:**
  - *Responsibilities:* Configures custom objects, fields, Record-Triggered Flows, Agentforce topics, and data access security controls.
  - *Touchpoints:* Salesforce Setup, Object Manager, Flow Builder, Prompt Builder, and Agent Builder.

1.3 Salesforce Architectural Stack

- **Custom Objects:** `Sender__c`, `Receiver__c`, `Parcel__c`, `Delivery__c`.
- **Standard Objects:** `Account`, `Contact`, `Case`, `Task`, `EmailMessage`, `User`.
- **Process Automation:** Record-Triggered Flows (status notifications), Auto-Launched Flows (Agentforce action invocations), and Scheduled Flows.
- **User Interface:** Custom `SwiftShip Tracker` Lightning App, Lightning Record Pages, Dynamic Forms, and Experience Cloud.

- **AI & Grounding:** Agentforce Agent Builder (autonomous parcel assistant) and Prompt Builder (grounded LLM prompts).

1.4 Data Model & Entity-Relationship Schema

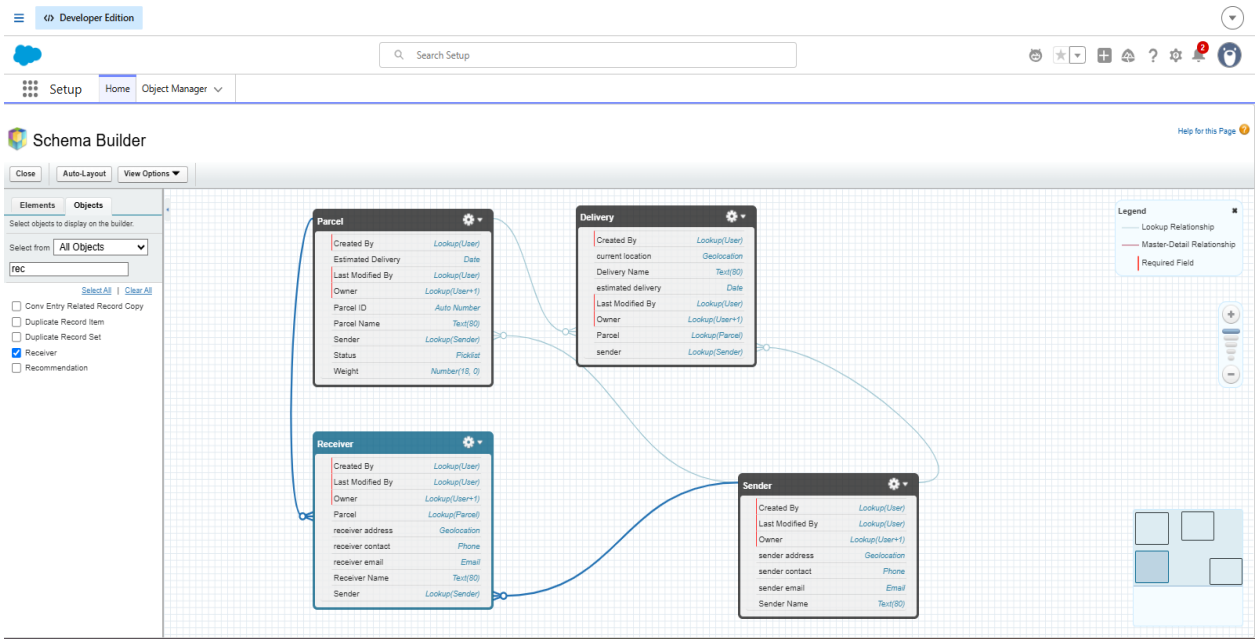
Object Name	API Name	Key Fields	Relationship Details
Sender	`Sender__c`	Name, Email, Phone, Street, City, State, Postal Code	Parent to `Parcel__c` (1:Many)
Receiver	`Receiver__c`	Name, Email, Phone, Delivery Address, Pincode	Parent to `Parcel__c` (1:Many)
Parcel	`Parcel__c`	Tracking ID (AutoNumber), Status, Weight, Est Delivery Date	Child of Sender/Receiver; Links to Delivery
Delivery	`Delivery__c`	Assigned Agent (User), Current Location, Delivery Date	Lookup associated with `Parcel__c` (1:1)

1.5 Security & Access Control Design

- **Organization-Wide Defaults (OWD):** Configured as **Private** for `Parcel__c` and `Delivery__c` to ensure couriers and customers cannot view global shipment records without explicit assignment or ownership.
- **Role Hierarchy:** Structured as:

  $$\text{System Administrator} \longrightarrow \text{Delivery Manager} \longrightarrow \text{Delivery Agent} \longrightarrow \text{Customer User}$$

- **Field-Level Security (FLS):** Personally Identifiable Information (PII) like phone numbers and physical coordinates are restricted to Administrator and Logistics Agent profiles only.
- **Sharing Rules:** Criteria-based sharing grants Read/Edit access to delivery agents only when `Delivery__c.Assigned_Agent__c` equals the current user record.


Developer Edition
Search Setup
Setup
Home
Object Manager
Schema Builder
Help for this Page
Close
Auto-Layout
View Options
Elements
Objects
Select objects to display on the builder.
Select from
All Objects
rec
Select All | Clear All
Conv Entry Related Record Copy
Duplicate Record Item
Duplicate Record Set
Receiver
Recommendation
Parcel
Created By Lookup(User)
Estimated Delivery Date
Last Modified By Lookup(User)
Owner Lookup(User+1)
Parcel ID Auto Number
Parcel Name Text(80)
Sender Lookup(Sender)
Status Picklist
Weight Number(18, 0)
Delivery
Created By Lookup(User)
current location Geolocation
Delivery Name Text(80)
estimated delivery Date
Last Modified By Lookup(User)
Owner Lookup(User+1)
Parcel Lookup(Parcel)
sender Lookup(Sender)
Receiver
Created By Lookup(User)
Last Modified By Lookup(User)
Owner Lookup(User+1)
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receiver address Geolocation
receiver contact Phone
receiver email Email
Receiver Name Text(80)
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sender address Geolocation
sender contact Phone
sender email Email
Sender Name Text(80)
Legend
Lookup Relationship
Master-Detail Relationship
Required Field

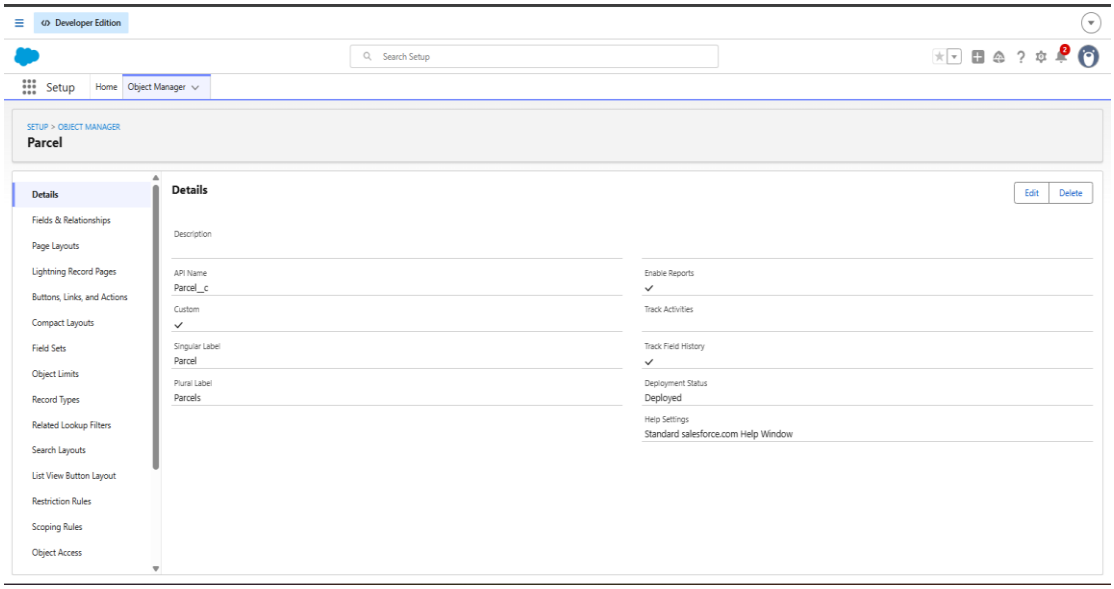
Phase 2: Salesforce Development – Backend & Configurations

2.1 Milestone 1: Salesforce Environment Setup

- Created and activated a dedicated Salesforce Developer Edition org with Einstein Generative AI and Agentforce runtime services enabled.
- Configured company settings, working business hours, standard currency, and security session controls.

2.2 Milestone 2 & 3: Custom Objects Creation & Navigation Tabs

- **`Sender__c`:** Built to store customer sender contact records, pick-up points, and account references.
- **`Receiver__c`:** Built to capture recipient names, verification contact numbers, and delivery instructions.
- **`Parcel__c`:** Configured with an Auto-Number identifier formatted as `TRK-{0000}` for tracking references.
- **`Delivery__c`:** Created to track delivery run assignments, real-time agent locations, and handover timestamps.
- Configured custom navigation tabs with custom Lightning icons, grouped neatly within the **SwiftShip Tracker** App launcher.



2.3 Milestone 4: Custom Fields & Relationship Cardinality

- **`Status__c` (Picklist on `Parcel__c`):** Values: `Booked`, `In Transit`, `Out for Delivery`, `Delivered`, `Delayed`.
- **`Weight__c` (Number on `Parcel__c`):** Decimal field storing parcel weight in kilograms for capacity planning.

- **`Est_Delivery_Date__c` (Date/Time on `Parcel__c`):** Estimated arrival timestamp generated upon booking.
- **`Current_Location__c` (Text on `Delivery__c`):** Checkpoint hub description updated during transit.
- **`Assigned_Agent__c` (Lookup on `Delivery__c`):** Direct reference linking the assigned courier `User` record.

2.4 Milestone 5: Prompt Builder Template Construction

- Built a custom Prompt Template grounded directly on live Salesforce database fields using merge syntax:

> *"You are the Swift Ship Tracker conversational agent. The customer is inquiring about shipment {!$Input:Parcel__c.Name}. Current status is {!$Input:Parcel__c.Status__c}, estimated delivery is {!$Input:Parcel__c.Est_Delivery_Date__c}, and latest location is {!$Input:Parcel__c.Delivery__r.Current_Location__c}. Provide a polite, concise update. If the parcel is marked Delayed, apologize and explain next steps. Do not invent details not present in the record.*

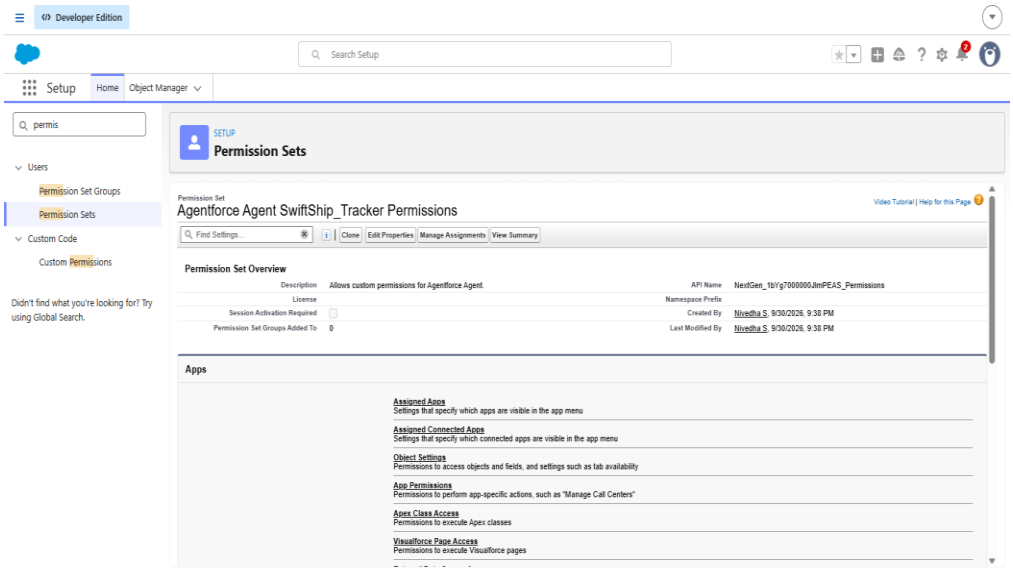
2.5 Milestone 6: Flow Automations & Business Logic

- **Record-Triggered Flow (`Parcel_Status_Change_Notification`):** Triggers when `Parcel__c.Status__c` is changed. Evaluates milestone logic and automatically fires an Email Alert to both sender and recipient.
- **Auto-Launched Flow (`SwiftShip_Query_Parcel_Status`):** Configured with an input variable `TrackingID` and an output record variable `ParcelRecord`. Serves as the callable programmatic action for the Agentforce conversational AI agent.

Phase 3: UI/UX Development & Customization

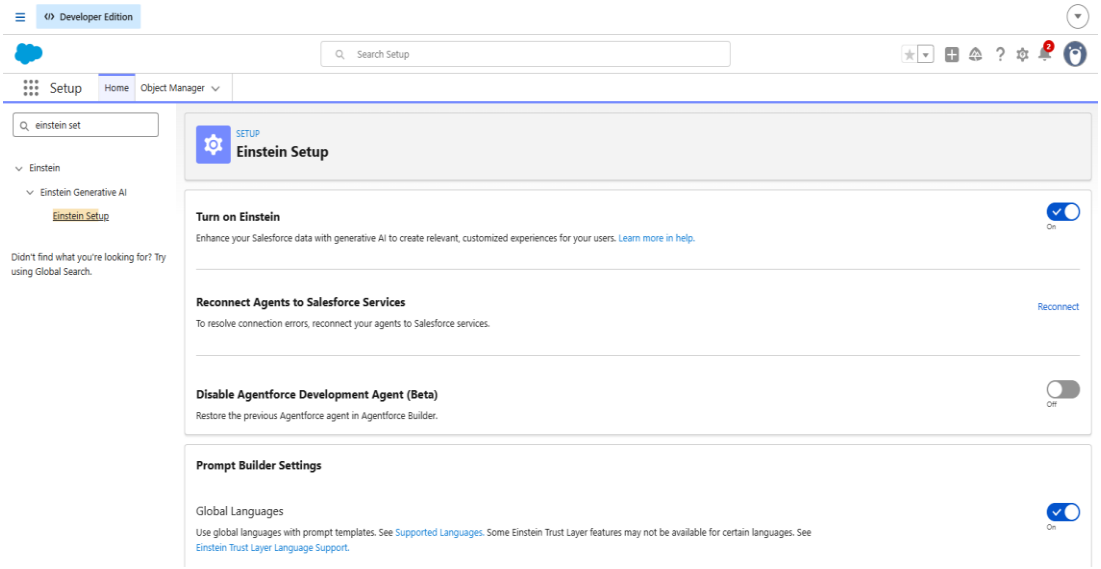
3.1 Milestone 1 & 2: Permission Sets Creation & Assignment

- **Permission Set (`Agentforce_Parcel_Permissions`):** Grants the background Agent user Read access to `Parcel__c`, `Delivery__c`, and prompt template execution permissions.
- **Permission Set (`Delivery_Agent_User_Permissions`):** Grants field couriers Read/Edit rights on parcels and deliveries assigned to them, while keeping administrative setup hidden.
- Assigned the permission sets to target users via **Setup $\rightarrow$ Users $\rightarrow$ Permission Set Assignments**.



3.2 Milestone 3: Org-Wide Activation for Agentforce

- Enabled generative AI services via **Setup $\rightarrow$ Einstein Setup $\rightarrow$ Turn on Einstein**.
- Enabled Agentforce features and validated Einstein Trust Layer guardrails (masking PII customer data before LLM execution).



3.3 Milestone 4: Building the Agent in Agent Builder

- Opened **Agent Builder** and created a new autonomous bot named **Parcel Tracking Agent**.
- Set the Agent Role: *"Support assistant dedicated to helping customers track parcel delivery statuses and get real-time shipping updates."*
- Defined the core Topic: **`SwiftShip Tracker`**.
- Associated the Auto-Launched Flow (`SwiftShip_Query_Parcel_Status`) as an active tool under the topic.

3.4 Milestone 5: Assigning Permissions to Agent User

- Verified that the underlying system service user associated with the Agent had the **Run Flows** permission enabled, along with Read access to custom object schemas and Prompt Builder endpoints.

Phase 4: Data Migration, Testing & Security

4.1 Verification Checklist & Testing Protocols

1. **Verify Agent Readiness:** Checked that the Parcel Tracking Agent was configured without syntax warnings and toggled to **Active** status in Agent Builder.
2. **Validate Assigned Topic:** Tested sample user phrases (*"Where is my package?"*, *"Track my parcel"*) to confirm the model consistently routes prompts to the `SwiftShip Tracker` topic.
3. **Check the Flow Association:** Confirmed that the agent parses the tracking identifier from user messages and passes it as an input parameter into the Auto-Launched Flow.
4. **Activate and Test the Agent:** Used the Agent Builder interactive preview console to simulate real customer conversations across various delivery stages.
5. **Validate Data in Salesforce:** Compared generated chat responses against live records in Salesforce Object Manager to verify absolute data fidelity with zero AI hallucinations.

4.2 Test Execution Results Matrix

Test Case	User Query Input	Expected Action	Observed Result	Status
TC-01	*"Where is my package TRK-1001?"*	Invoke topic, run Flow, return live location & status	Returned "In Transit - Central Hub" with ETA	**Passed**
TC-02	*"Is TRK-1002 delivered?"*	Query `Parcel__c`, evaluate `Delivered` status	Confirmed successful delivery and handover time	**Passed**
TC-03	*"Status of TRK-9999 (invalid)"*	Flow returns null; Agent runs fallback response	Responded politely asking to re-check tracking ID	**Passed**
TC-04	*"Change delivery address via chat"*	Deny action; restrict write operations per security policy	Advised user to contact customer support hotline	**Passed**

4.3 Testing Scope & Production Boundary Statement

All functional testing was executed inside a standalone **Salesforce Developer Edition Org** to validate topic recognition, Flow execution, and prompt grounding fidelity. Production-grade multi-tier sandbox deployments (Dev $\rightarrow$ QA $\rightarrow$ UAT $\rightarrow$ Staging $\rightarrow$ Production), enterprise CI/CD release pipelines (Salesforce DX / GitHub Actions), and high-volume load testing are outside the scope of this implementation.

Phase 5: Deployment, Documentation & Maintenance

5.1 Maintenance, Monitoring & Troubleshooting Playbook

- **Conversational Audit Logs:** Periodically review Agentforce session transcripts to check intent-classification accuracy and identify unrecognized customer phrasing.
- **Flow Exception Debugging:** Review automated Flow Fault emails in Setup and monitor paused flow interviews to catch missing input variables or validation rule blocks.
- **Prompt Grounding Refinement:** Adjust Prompt Builder templates as new shipping statuses or delivery exceptions are added to the CRM.

5.2 Project Documentation & Artifact Repository

- **Technical Architecture Documents:** Full object relationship schemas, field definition tables, and Flow logic diagrams stored in the GitHub repository.
- **Agent Operations Runbook:** Clear procedural guides instructing dispatchers and logistics agents how to update statuses and use Lightning Console pages.

5.3 Project Demonstration Video Outline

1. **Introduction (1.5 mins):** Business background, parcel delivery challenges, and project solution scope.
2. **CRM Backend Tour (3 mins):** Walkthrough of `Parcel__c`, `Delivery__c`, custom tabs, status picklists, and relationship lookups.
3. **Automations & AI Setup (2.5 mins):** Review of the Record-Triggered notification Flow, Auto-Launched Flow, and Prompt Builder grounded template.
4. **Live Agentforce Demonstration (3 mins):** Interactive live test in Agent Builder showing tracking queries, dynamic Flow execution, and real-time database validation.

5.4 Conclusion

The **Swift Ship Tracker** project demonstrates a complete Salesforce implementation that combines declarative CRM architecture with autonomous generative AI agents. By automating parcel status tracking and delegating routine inquiries to Agentforce, the solution eliminates customer support bottlenecks, provides real-time shipment transparency, and streamlines field delivery operations.